Revisions

Rev.	Date	Author	Description
A	15/11/2020	Fabrice Pichard	Created from FBA IND-G (Sevcon track inverters) Adapted for 12V Zapi track inverters : 12V supply to track inverters, dedicated 24V supply to variators for brake release solenoid
B	27/11/2020	P.L.P	Relay Batt changing / Add wire from battery_350V to permanent_12V
C	13/08/2021	E.P	Précision schéma Fusebox
D	29/11/2021	E.P	Maj Faisceau complet + Modif longueur

Splice_12V

From	To	Conductor	Color	Gauge
Splice_12V	FuseBox_BA.A6	W6.Red		1,5 mm²
Splice_12V	FuseBox_BA.A8	W4.Red		1,5 mm²
Splice_12V	FuseBox_BA.A10	W90.Red		1,5 mm²
Splice_12V	FuseBox_BA.B2	W1.Red		1,5 mm²
Splice_12V	FuseBox_BA.C1	W2.Red		1,5 mm²
Splice_12V	Supply.A	W51.Red		4 mm²

Splice_0V

From	To	Conductor	Color	Gauge
Splice_0V	24V_Converter_10807.B	W79.Black		1,5 mm²
Splice_0V	24V_Converter_10807.D	W80.Black		1,5 mm²
Splice_0V	Battery_350V.B	W78.Black		1,5 mm²
Splice_0V	Charger.B	W84.Black		1,5 mm²
Splice_0V	Controler_Left.B	W18.Black		1,5 mm²
Splice_0V	Controlers_Right_Pump.B	W82.Black		1,5 mm²
Splice_0V	Controlers_Right_Pump.F	W23.Black		1 mm²
Splice_0V	Controlers_Right_Pump.G	W9.Black		1 mm²
Splice_0V	Emergency_Brake_Left.B	W26.Black		1,5 mm²
Splice_0V	Emergency_Brake_Right.B	W83.Black		1,5 mm²
Splice_0V	FuseBox_BA.B3	W81.Black		1,5 mm²
Splice_0V	Supply.B	W52.Black		4 mm²

FuseBox_BA

Cavity	To	Conductor	Color	Gauge
A1	FuseBox_BA.A4	W5.Red		1,5 mm²
A3	MC42_Cmd.C	W17.Violet		1 mm²
A4	FuseBox_BA.A1	W5.Red		1,5 mm²
A5	Battery_350V.A	W24.Yellow		1,5 mm²
A6	Splice_12V	W6.Red		1,5 mm²
A7	Charger.A	W88.Red		1,5 mm²
A8	Splice_12V	W4.Red		1,5 mm²
A9	Battery_350V.C	W25.Gray		1,5 mm²
A10	Splice_12V	W90.Red		1,5 mm²
A11	Splice_Emergency_Brake	W92.Red		1,5 mm²
B1	MC42_Cmd.B	W55.Orange		1 mm²
B2	Splice_12V	W1.Red		1,5 mm²
B3	Splice_0V	W81.Black		1,5 mm²
B4	Permanent_12V	W28.Red		1 mm²
B5	Battery_350V.D	W27.Red		1 mm²
B9	Splice_12V_Controller	W30.Red		1,5 mm²
B10	24V_Converter_10807.A	W7.Gray		1,5 mm²
B11	24V_Converter_10807.C	W13.Green		1 mm²
B12	Splice_24V_Plus	W32.Yellow		1,5 mm²
C1	Splice_12V	W2.Red		1,5 mm²
C3	MC42_Cmd.A	W15.Green		1 mm²
C4	Splice_12V_Controller	W89.Red		1,5 mm²
C5	Controlers_Right_Pump.A	W10.Green		1 mm²
C6	Splice_12V_Controller	W29.Red		1,5 mm²
C7	Controler_Left.A	W12.Green		1 mm²
C8	Splice_12V_Controller	W110.Red		1,5 mm²
C9	Controlers_Right_Pump.E	W91.Red		1,5 mm²
D1	MC42_Cmd.D	W56.Orange		1 mm²
D3	Splice_12V_Controller	W3.Red		1,5 mm²

Permanent_12V

From	To	Conductor	Color	Gauge
Permanent_12V	FuseBox_BA.B4	W28.Red		1 mm²

Splice_12V_Controller

From	To	Conductor	Color	Gauge
Splice_12V_Controller	FuseBox_BA.B9	W30.Red		1,5 mm²
Splice_12V_Controller	FuseBox_BA.C4	W89.Red		1,5 mm²
Splice_12V_Controller	FuseBox_BA.C6	W29.Red		1,5 mm²
Splice_12V_Controller	FuseBox_BA.C8	W110.Red		1,5 mm²
Splice_12V_Controller	FuseBox_BA.D3	W3.Red		1,5 mm²

Splice_24V_Plus

From	To	Conductor	Color	Gauge
Splice_24V_Plus	Controler_Left.E	W8.Gray		1,5 mm²
Splice_24V_Plus	Controlers_Right_Pump.H	W35.Gray		1,5 mm²
Splice_24V_Plus	FuseBox_BA.B12	W32.Yellow		1,5 mm²

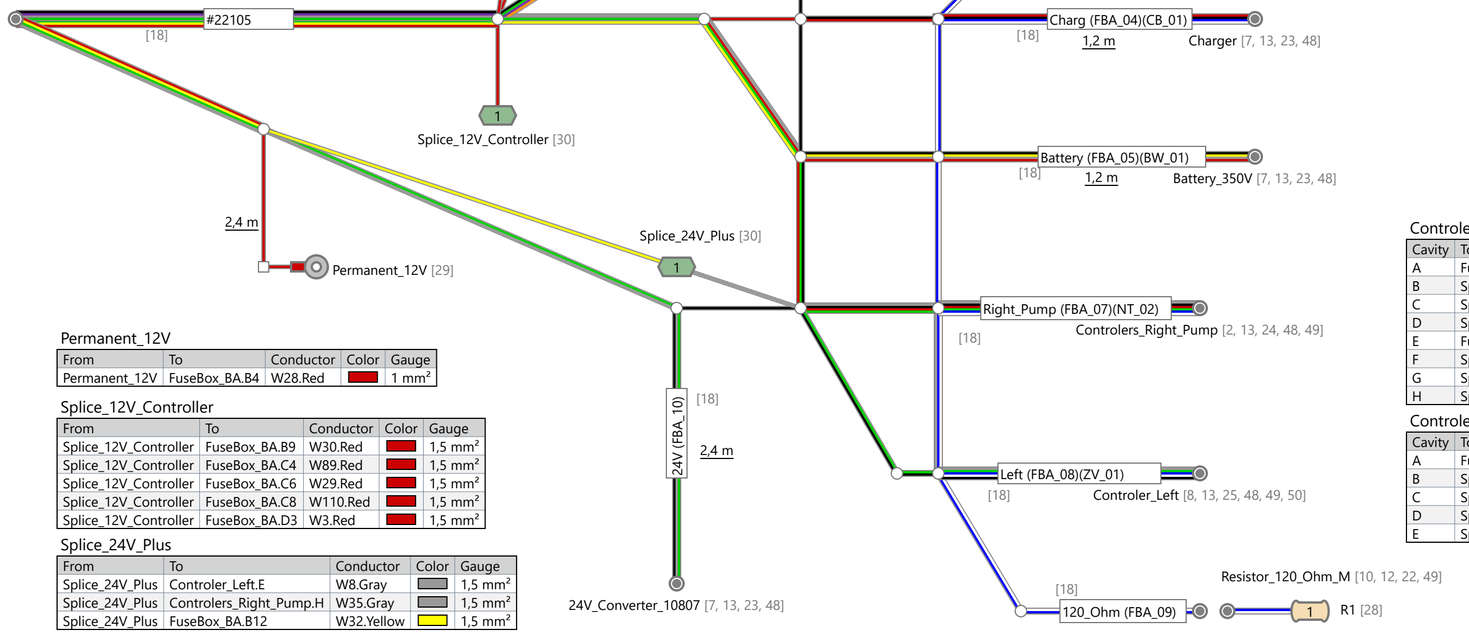
24V_Converter_10807

Cavity	To	Conductor	Color	Gauge
A	FuseBox_BA.B10	W7.Gray		1,5 mm²
B	Splice_0V	W79.Black		1,5 mm²
C	FuseBox_BA.B11	W13.Green		1 mm²
D	Splice_0V	W80.Black		1,5 mm²

Splice_Can_Low, Splice_Can_High

From	To	Conductor	Color	Gauge
Splice_Can_High	CanBus_From_MC42.A	W94.White		1 mm²
Splice_Can_High	Charger.C	W96.White		1 mm²
Splice_Can_High	Controler_Left.C	W19.White		1 mm²
Splice_Can_High	Controlers_Right_Pump.C	W99.White		1 mm²
Splice_Can_High	Resistor_120_Ohm_F.A	W95.White		1 mm²
Splice_Can_Low	CanBus_From_MC42.B	W106.Blue		1 mm²
Splice_Can_Low	Charger.D	W102.Blue		1 mm²
Splice_Can_Low	Controler_Left.D	W21.Blue		1 mm²
Splice_Can_Low	Controlers_Right_Pump.D	W105.Blue		1 mm²
Splice_Can_Low	Resistor_120_Ohm_F.B	W107.Blue		1 mm²





Emergency_Brake_Left

From	To	Conductor	Color	Gauge
Emergency_Brake_Left.A	Splice_Emergency_Brake	W93.Red		1,5 mm²
Emergency_Brake_Left.B	Splice_0V	W26.Black		1,5 mm²

Emergency_Brake_Right

From	To	Conductor	Color	Gauge
Emergency_Brake_Right.A	Splice_Emergency_Brake	W108.Red		1,5 mm²
Emergency_Brake_Right.B	Splice_0V	W83.Black		1,5 mm²

Supply

From	Cavity	To	Conductor	Color	Gauge
Supply.A	A	Splice_12V	W51.Red		4 mm²
Supply.B	B	Splice_0V	W52.Black		4 mm²

MC42_Cmd

From	To	Conductor	Color	Gauge
MC42_Cmd.A	FuseBox_BA.C3	W15.Green		1 mm²
MC42_Cmd.B	FuseBox_BA.B1	W55.Orange		1 mm²
MC42_Cmd.C	FuseBox_BA.A3	W17.Violet		1 mm²
MC42_Cmd.D	FuseBox_BA.D1	W56.Orange		1 mm²

CanBus_From_MC42

Cavity	To	Conductor	Color	Gauge
A	Splice_Can_High	W94.White		1 mm²
B	Splice_Can_Low	W106.Blue		1 mm²

Modification cable blindé
rouge sur blanc
vert sur bleue

Charger

Cavity	To	Conductor	Color	Gauge
A	FuseBox_BA.A7	W88.Red		1,5 mm²
B	Splice_0V	W84.Black		1,5 mm²
C	Splice_Can_High	W96.White		1 mm²
D	Splice_Can_Low	W102.Blue		1 mm²

Battery_350V

Cavity	To	Conductor	Color	Gauge
A	FuseBox_BA.A5	W24.Yellow		1,5 mm²
B	Splice_0V	W78.Black		1,5 mm²
C	FuseBox_BA.A9	W25.Gray		1,5 mm²
D	FuseBox_BA.B5	W27.Red		1 mm²

Controlers_Right_Pump

Cavity	To	Conductor	Color	Gauge	Notes
A	FuseBox_BA.C5	W10.Green		1 mm²	Controler_1_Fuse_20A
B	Splice_0V	W82.Black		1,5 mm²	
C	Splice_Can_High	W99.White		1 mm²	
D	Splice_Can_Low	W105.Blue		1 mm²	
E	FuseBox_BA.C9	W91.Red		1,5 mm²	Controler_3_Fuse_20A
F	Splice_0V	W23.Black		1 mm²	
G	Splice_0V	W9.Black		1 mm²	CanBus shield grounding
H	Splice_24V_Plus	W35.Gray		1,5 mm²	

Controler_Left

Cavity	To	Conductor	Color	Gauge
A	FuseBox_BA.C7	W12.Green		1 mm²
B	Splice_0V	W18.Black		1,5 mm²
C	Splice_Can_High	W19.White		1 mm²
D	Splice_Can_Low	W21.Blue		1 mm²
E	Splice_24V_Plus	W8.Gray		1,5 mm²

Resistor_120_Ohm_M

Cavity	To	Conductor	Color	Gauge
A	R1.POS	W16.White		1 mm²
B	R1.NEG	W14.Blue		1 mm²

Resistor_120_Ohm_F

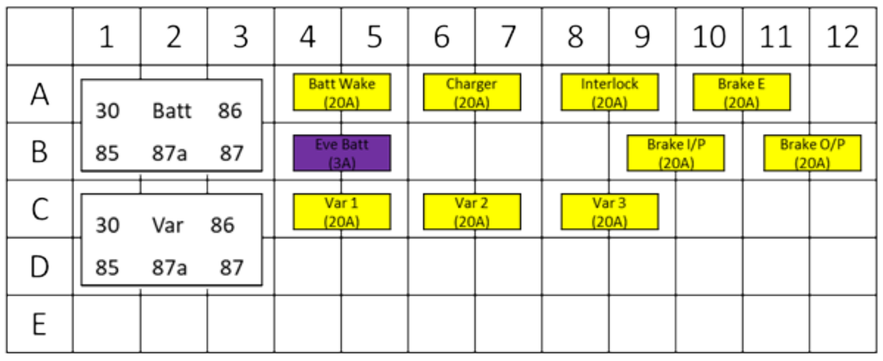
Cavity	To	Conductor	Color	Gauge
A	Splice_Can_High	W95.White		1 mm²
B	Splice_Can_Low	W107.Blue		1 mm²

R1

From	To	Conductor	Color	Gauge
R1.NEG	Resistor_120_Ohm_M.B	W14.Blue		1 mm²
R1.POS	Resistor_120_Ohm_M.A	W16.White		1 mm²

A - Input + C - Output +
B - Input - D - Output -

On Sevcon equiped Commuters, Sevcon inverters are supplied on 24V with independant 24V ground.
On Zapi equiped Commuters, Zapi inverters are supplied on 12V and used to pull 24V brake release coils to ground.
So on Zapi equiped Commuters, 12V and 24V supplies have common ground.



	Name	Date
Drawn By	----	----
Eng Approved	----	----
Mfg Approved	----	----
QA	----	----

Title: #30202 - FuseBox_Electric_Drive Zapi (FBA) - IND D - FR

Description

Revision: ----

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Title:	#30202 - FuseBox_Electric_Drive Zapi (FBA) - IND D - FR
Description:	

Id	Type	Part Number	Quantity	Description	Manufacturer
1	Connector	#26535 - Littelfuse_280_60_way_FuseBox	1	Littelfuse PDM21001LXM	Littelfuse
2	Connector	#13000 - Delphi_150_8_way_female	1	Delphi 12047937	Delphi
3	Connector	#14869 - Delphi_150_2_way_female	2	Delphi 12052641	Delphi
4	Connector	#16777 - Delphi_280_2_way_male	2	Delphi 15300002	Delphi
5	Connector	#17183 - Delphi_480_2_way_male	1	Delphi 12065863	Delphi
6	Connector	#26525 - Delphi_150_4_way_male	1	Delphi 12160825	Delphi
7	Connector	#28050 - Delphi_150_4_way_female	3	Delphi 12160482-B	Delphi
8	Connector	#15960 - Delphi_150_6_way_female	1	Delphi 12052848	Delphi
9	Connector	#17351 - Delphi_280_2_way_female	2	Delphi 15300027	Delphi
10	Connector	#15215 - Delphi_150_2_way_male	1	Delphi 12162000	Delphi
11	Contact	#17237 Delphi_480_contact_male 5mm² 3.49-5.24mm	2	Delphi 12052172	Delphi
12	Contact	#10019 Delphi_150_contact_male 0.5-1.0mm² 1.69-2.78mm	6	Delphi 12045773	Delphi
13	Contact	#13657 Delphi_150_contact_female 0.5-1.0mm² 2.03-2.80mm	29	Delphi 12048074	Delphi
14	Contact	#13581 Delphi_280_contact_male 0.5-0.8mm² 1.69-2.78mm	4	Delphi 12048159	Delphi
15	Contact	#14173 Delphi_280_contact_female 2.0-3.0mm² 2.45-3.80mm	29	Delphi 12129493	Delphi
16	Fuse	#37333 Fuse 3A Mini	1	Littelfuse 0297003.H	Littelfuse
17	Fuse	#20949 Fuse 20A Mini	9	Littelfuse 0297020.WXNV	Littelfuse
18	Label	Generic White Label	13	Generic Label	
19	Lock	#18797 - Delphi_280_2_way_lock	4	Delphi 15300014-B	Delphi
20	Lock	#10273 - Delphi_480_2_way_lock	1	Delphi 12059897	Delphi
21	Lock	#23010 - Delphi_150_4_way_male_lock	1	Delphi 12176368	Delphi
22	Lock	#19468 - Delphi_150_2_way_lock	3	Delphi 12052634	Delphi
23	Lock	#22024 - Delphi_150_4_way_female_lock	3	Delphi 12160480	Delphi
24	Lock	#12834 - Delphi_150_8_way_female_lock	1	Delphi 15324525	Delphi
25	Lock	#14848 - Delphi_150_6_way_lock	1	Delphi 12052850	Delphi
26	Relay	#25661 - Relais Micro 280 Automotive - 4 PIN	1		Mouser
27	Relay	#36314 5 PIN SPDT sealed relay w/ suppress.diode Micro 280	1	Song Chuan 3011CSD112VDC	Song Chuan
28	Resistor	#27646 120Ω ±1%, 0.6W metal film resistor	1	TE Connectivity LR1F120R	TE Connectivity
29	Ring Terminal	16 AWG Red	1	Generic Ring Terminal	
30	Splice	Generic Splice	7	Generic Splice	
31	Wire	1 mm² - Orange	≥ 4,8 m		
32	Wire	1 mm² Green	≥ 4,8 m	Generic Wire	
33	Wire	1 mm² Violet	≥ 2,4 m	Generic Wire	
34	Wire	1.5 mm² - Yellow	≥ 1,2 m		
35	Wire	1.5 mm² Black	≥ 1,9 m	Generic Wire	
36	Wire	1.5 mm² Gray	≥ 1,2 m	Generic Wire	
37	Wire	1.5mm² - Black	≥ 10,1 m		
38	Wire	1.5mm² - Red	≥ 6 m		
39	Wire	1mm² - Blue	≥ 3,6 m		
40	Wire	1mm² - White	≥ 3,6 m		
41	Wire	1mm² Black			
42	Wire	4 mm² - Black	1,4 m		
43	Wire	4 mm² - Red	≥ 1,4 m		
44	Wire	1.5mm² - Gray	≥ 2,4 m		
45	Wire	1mm² Red	≥ 3,6 m		
46	Wire Seal	#19508 - Delphi 280 Cavity plug	35	Delphi 12010300	Delphi
47	Wire Seal	#15414 - Delphi 480 Cable seal – Red – 2.81-3.76	2	Delphi 12052452	Delphi
48	Wire Seal	#17898 - Delphi 150 Cable seal – White – 1.60-2.15	25	Delphi 12089442	Delphi
49	Wire Seal	#14820 - Delphi 150 Cable seal – Blue – 1.29-1.70	10	Delphi 12052925	Delphi
50	Wire Seal	#19593 - Delphi 150 Cavity plug	1	Delphi 12059168	Delphi
51	Wire Seal	#13381 - Delphi 280 Cable seal – Green– 2.03-2.85	33	Delphi 12015284	Delphi